

DLA Draft

Developmental Learning Architecture: Learning History, Fast-Weight Traces, and Future Adaptation

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Draft v0.2 (validation-stage) — workshop/arXiv

Abstract

A conventional neural learner is treated as a fixed function trained once by an external optimizer. This paper studies a more developmental question: does a learner's prior learning history change how it will learn in the future, and which internal state carries this effect? We propose a Developmental Learning Architecture (DLA) that keeps standard MLP/Transformer backbones unchanged but augments each weight with per-parameter fast weights, plasticity, and sleep consolidation. In a 6.59M Chinese character GPT we find that different curriculum histories lead to different future adaptation on an unseen domain, and that this effect is causally tied to the fast-weight state w_{fast} : replacing a hard→easy learner's fast weights with an easy→hard learner's significantly degrades future adaptation ($n=12$, gain@40 difference -0.019 , bootstrap CI excludes zero, Cohen's $d \approx -0.7$). Norm-matching and shuffling do not remove the effect, so it is not explained by weight magnitude or random parameter noise. The effect replicates on a second Shakespeare character-level backbone. We do not find evidence that more experience monotonically accelerates learning: multiple controlled longitudinal studies return null results. The conclusion is that learning

history leaves a persistent, partially localizable trace in transient learner state, with w_{fast} providing a causal component of history-dependent future adaptation.

1. Introduction

1.1 What problem exists

Modern neural networks are trained and then frozen. When they learn a second task, they either overwrite the first (catastrophic forgetting) or treat the two tasks as independent. More fundamentally, standard training does not allow the model itself to become a different kind of learner: the learning algorithm is fixed outside the model.

We ask a problem that sits before “continual learning” and “meta-learning”:

Can what a model has experienced change how it will learn in the future, and which part of the model carries that developmental change?

1.2 Why it matters

If learning history only changes stored knowledge, the learner remains a static function with a growing database. If learning history also changes the learner state, then a small model could, in principle, become a better learner through experience—without changing its architecture. This is the key motivation for “growing models” and for understanding whether development is a meaningful object in machine learning.

The practical significance is twofold. First, it tells us which parts of a model to preserve or transfer between learning phases. Second, it gives a falsifiable framework for claims such as “more experience makes models learn faster” — a claim we test and do not find support for.

2. How existing work addresses the problem

- **Catastrophic forgetting methods** (regularization, replay, EWC) address memory, not the change of the learner itself.
- **Fast-weight models** add a temporary memory but usually keep the learning rule external.

- **Meta-learning** optimizes a learning rule over many episodes; it asks whether a rule can be learned, but rarely whether one individual's rule develops within a single lifetime.
 - **Learned plasticity** (differentiable plasticity, ANML) shows that plasticity can be meta-learned, but does not separate *body state* from *learning-rule parameters* as causal carriers.
 - **We therefore design DLA** to ask a mechanistic question that is usually skipped: given the same task sequence, which state component carries history-dependent future adaptation?
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3. Our solution: DLA

3.1 Design principle

We do not modify the Transformer skeleton. Instead, each weight matrix is accompanied by:

- W_{slow} : long-term consolidated knowledge
- W_{fast} : a fast learning trace
- P : per-parameter plasticity
- Q : sleep eligibility trace

Effective weights:

$$W_{\text{eff}} = W_{\text{slow}} + \text{softplus}(P) * W_{\text{fast}}$$

3.2 Wake and sleep

During learning, W_{fast} is updated with Adam-style moments, gated by $\text{softplus}(P)$. At task boundaries (sleep), part of W_{fast} is consolidated into W_{slow} through Q , and W_{fast} is decayed. This is the fast/slow separation. **Algorithm 1: DLA wake-sleep cycle**

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1: for task t in curriculum do
2:   for step in 1..T do
3:     compute  $g = dL/dW_{\text{eff}}$ 
4:     update  $W_{\text{fast}}$  with Adam( $g$ ), gated by  $\text{softplus}(P)$ 
5:     update  $P$  from progress/relevance
6:   end for
7:   consolidate:  $W_{\text{slow}} += \beta * Q + \gamma * W_{\text{fast}}$ 
8:   decay:  $W_{\text{fast}} *= c$ ,  $Q *= d$ 
9: end for

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3.3 Experimental design to isolate development

To test whether history changes the future learner, we:

1. Build two histories with the same three domains but different order:
 - easy→hard (EH)
 - hard→easy (HE)
 2. Probe each individual on a never–seen domain D.
 3. Perform body/component cross–injection to see which state component transfers the history effect.
 4. Run norm–matched, shuffled, and module–localized controls to rule out magnitude/randomness.
 5. Compare with standard baselines (AdamW, replay, EWC).
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4. Experimental setup

Primary backbone: 6.59M Chinese character GPT (6 layers, 8 heads, 256 dim, vocab 7280), pretrained on Chinese Wikipedia. Curriculum domains: Wikipedia, SFT–style QA, science Wikipedia; unseen domain D is a science slice. Second setting: ~10.65M Shakespeare character GPT (vocab 65). Metrics: gain@40, LE_D, T80, paired bootstrap CI, sign test.

Table 1: Common experimental hyperparameters.

Hyperparameter	Value
Block size	128
Batch size	32
Steps per task	40
AdamW learning rate (baselines/DLA base)	1e−4
EWC lambda	1e3
Replay buffer capacity	48 batches
Seeds (primary setting)	12
Seeds (baselines)	5
Seeds (second setting)	2

5. Results: how well does it work

5.1 Fast/slow separation protects consolidated memory

DLA matches tuned AdamW on new-domain adaptation (gain +14.0% vs +13.0%) while slow-memory forgetting is ≈ 0 (−0.4%). This shows the mechanism does not sacrifice memory for plasticity.

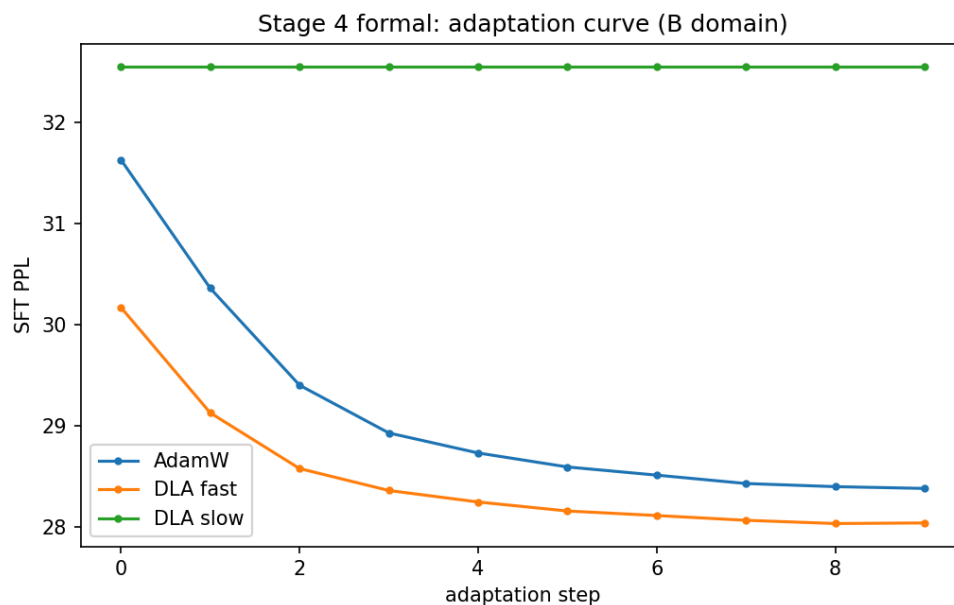
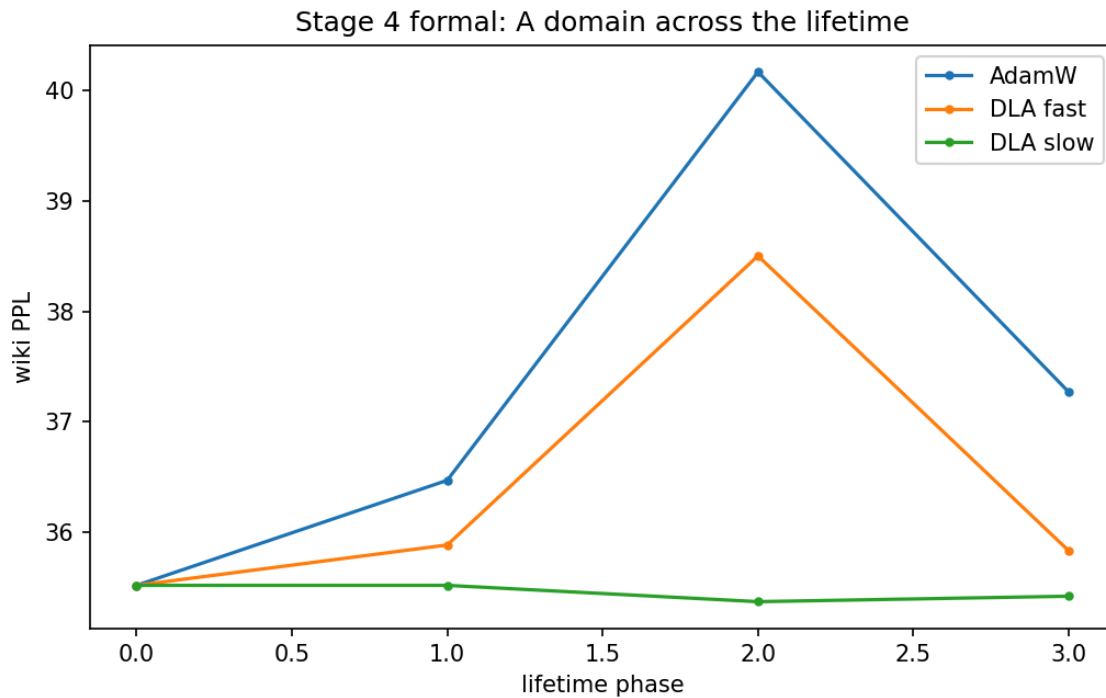


Figure 1b: Adaptation curve on the new domain.

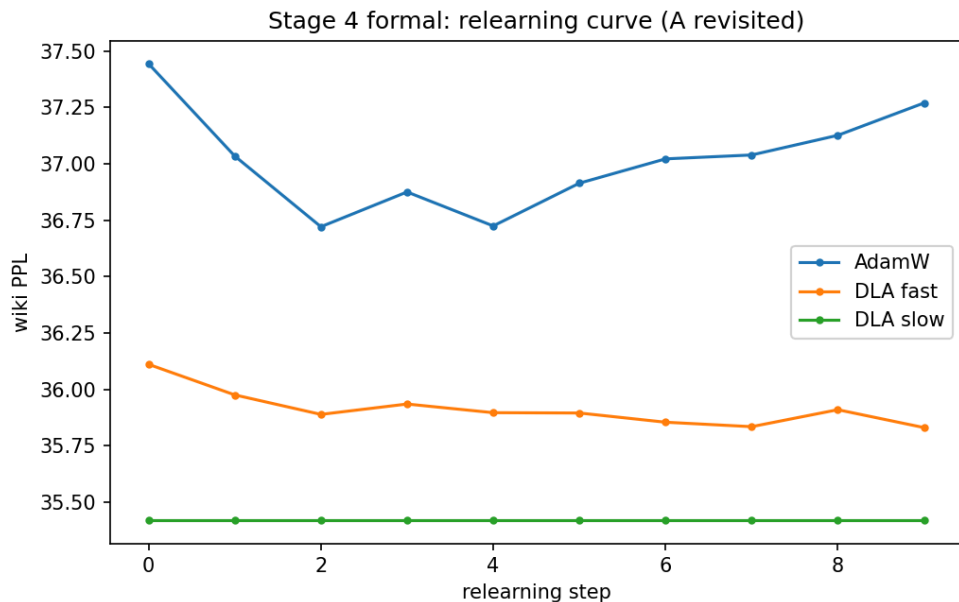


Figure 1c: Relearning curve.

5.2 Learning history changes future adaptation

Across histories, HE learners adapt better to unseen D than EH learners. This is not unique to DLA; but DLA is most robust after a difficult history.

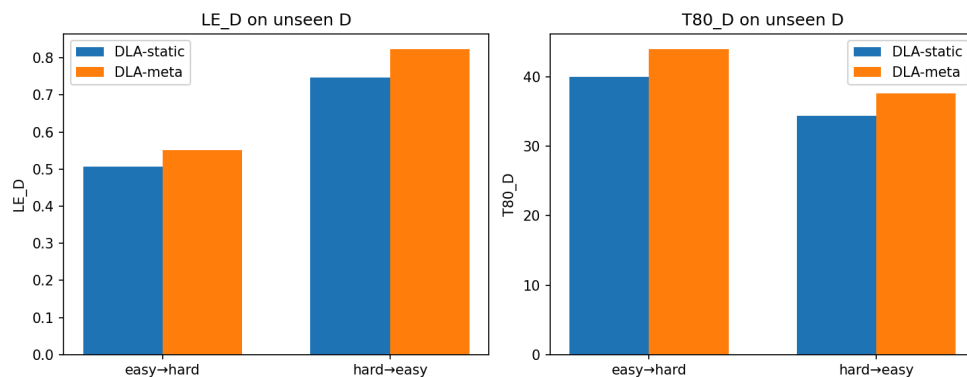


Figure 2: History effect on unseen D (2x2 development experiment).

5.3 The effect is not carried by the learning-rule parameters ϕ

Body $\times\phi$ cross-injection shows that swapping ϕ between histories changes future adaptation much less than swapping body state. ϕ is not a stable/dominant carrier in this setting.

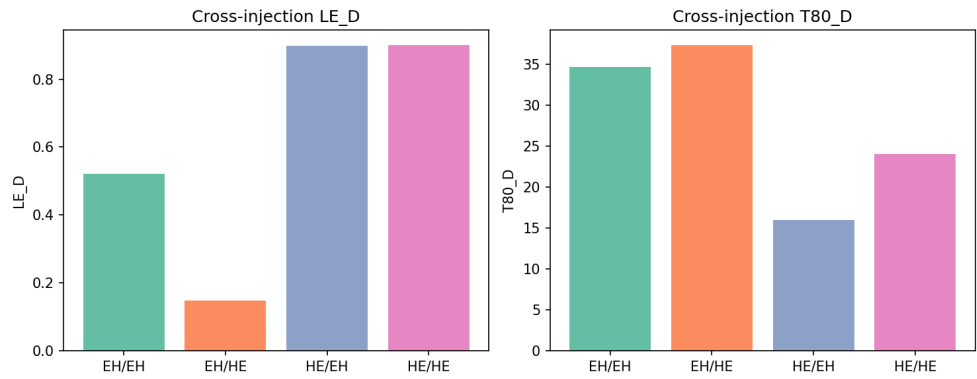


Figure 3: Body $\times$ ϕ causal dissection.

5.4 W_{fast} is a causal component

Core result (n=12):

condition	mean gain@40
HE/HE	+0.014
HE + EH W_{fast}	-0.005

Paired HE $\rightarrow$ raw EH: mean -0.019 , 95% CI $[-0.033, -0.005]$, Cohen’s $d \approx -0.71$, sign $p=0.019$ (10/12 negative). Initial PPL differences are small; the effect appears in the adaptation trajectory.

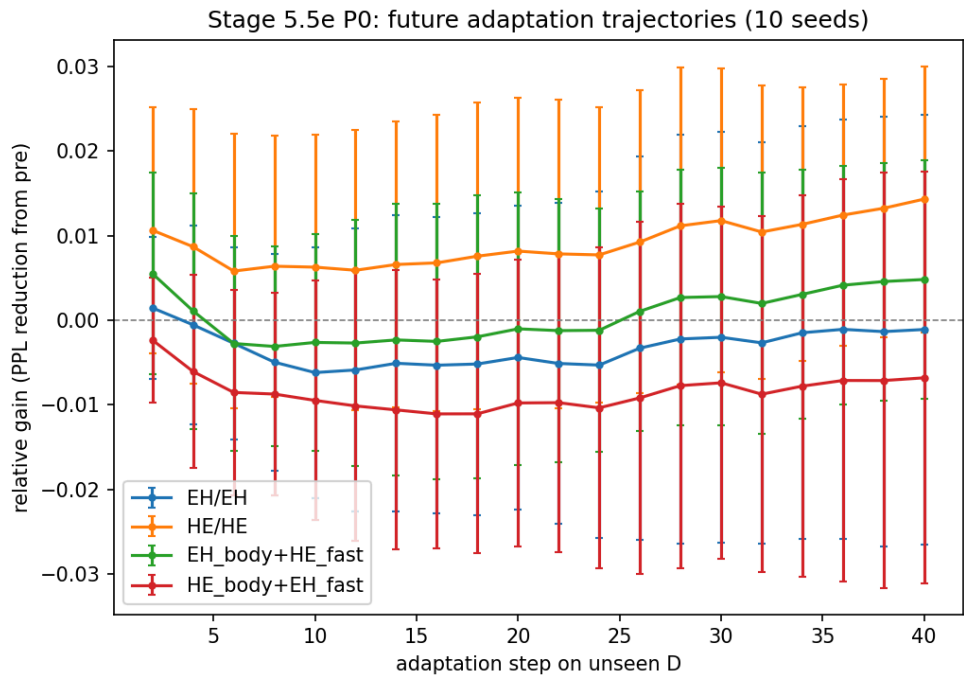


Figure 4: P0 future–adaptation trajectories.

5.5 Controls: magnitude, randomness, module

control	mean Δ gain@40	95% CI	d
raw EH W_fast	-0.019	[-0.034, -0.005]	-0.71
norm-matched EH W_fast	-0.022	[-0.034, -0.009]	-0.90
shuffled EH W_fast	-0.017	[-0.023, -0.009]	-1.25
EH embedding W_fast	-0.006	[-0.009, -0.002]	-0.87
EH attention W_fast	-0.012	[-0.018, -0.006]	-1.06
EH MLP W_fast	-0.017	[-0.025, -0.008]	-1.05

Norm-matching does not rescue the effect; shuffling also does not remove it. Module localization is suggestive of MLP and attention, with embedding weaker.

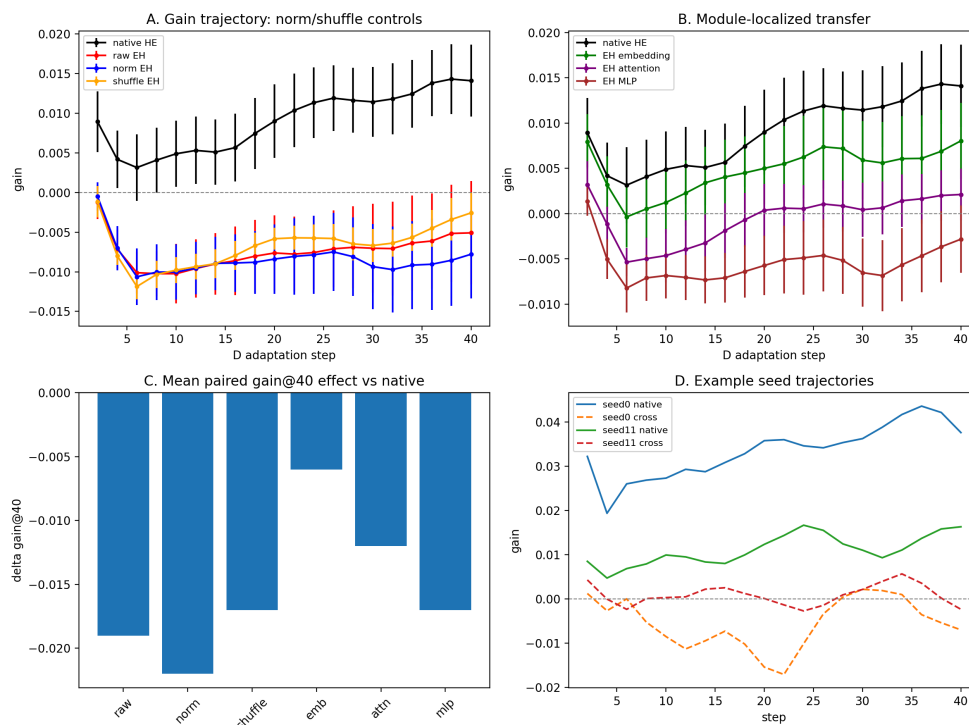


Figure 5: W_fast cross-injection and full future-adaptation trajectories. A: norm/shuffle controls. B: module-localized transfer. C: paired gain@40 effects. D: example trajectories.

5.6 Fair baselines

method	EH	HE	HE-EH
AdamW	0.339	0.675	+0.336
AdamW+replay	0.375	0.961	+0.586
EWC	0.292	0.895	+0.603

method	EH	HE	HE-EH
DLA	0.517	0.709	+0.192

DLA is the most robust after an easy→hard history; standard learners show larger HE performance but larger EH degradation.

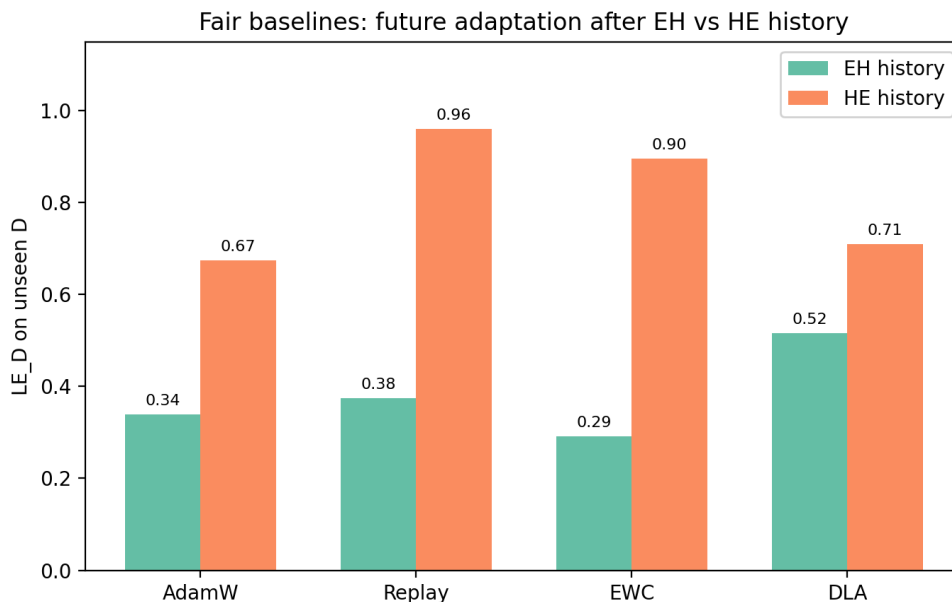


Figure 6: Fair baselines — future adaptation after EH vs HE history.

5.7 Second-setting replication

Shakespeare char GPT, 2 seeds: HE/HE gain@40 +0.052/+0.042; HE+EH fast +0.004/−0.001. The destructive effect replicates in direction.

5.8 Negative result: more experience does not necessarily make learning faster

Stage 5 (difficulty-normalized): flat LE. Stage 6 (matched difficulty): $p=0.17$. Stage 7 (cross-domain, 20 seeds): $p=0.82$. Stage 8 (physics near-transfer, 20 seeds): $d=-0.17$. We therefore do not claim “more experience → faster learning”.

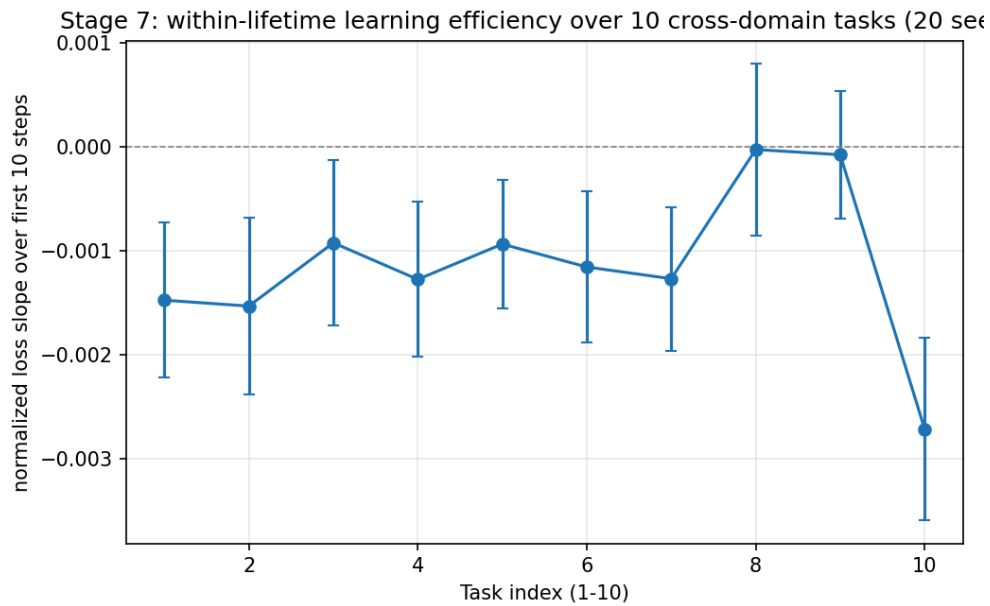


Figure 7a: Negative result — cross-domain longitudinal (Stage 7, 20 seeds).

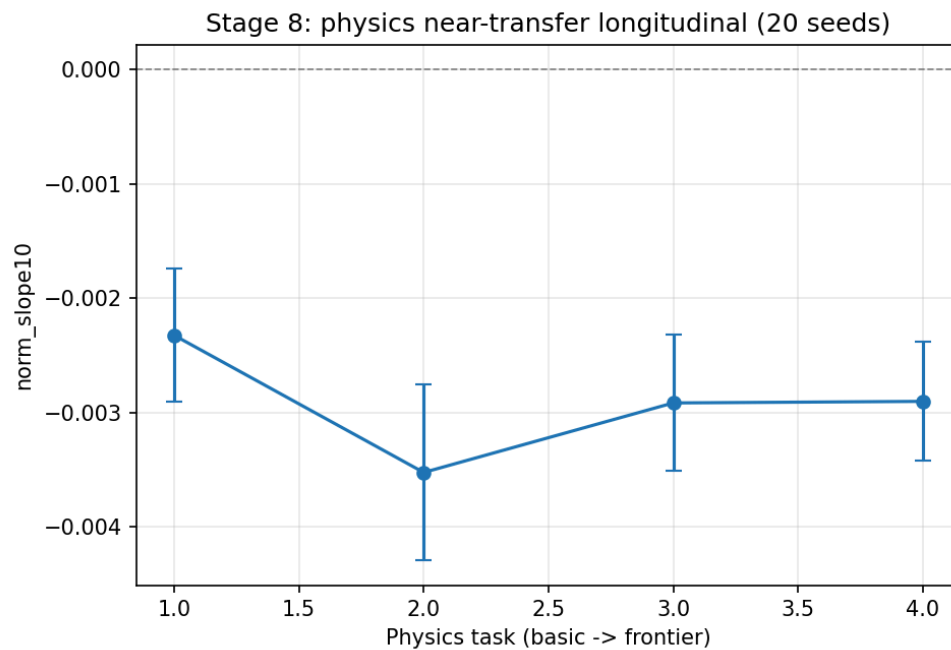


Figure 7b: Negative result — physics near-transfer (Stage 8, 20 seeds).

6. Limitations

- Second-setting $n=2$; baselines $n=5$.
- Single small backbones; no large-scale validation.
- Sleep consolidation writes only a small amount into w_{slow} ; fast/slow separation is mostly isolation.

- No pre-registration; we report $n=10 \rightarrow n=12$ transparently.
 - We cannot yet claim W_{fast} is the only carrier or that module localization is definitive.
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7. Conclusion and significance

The learner is not fully characterized by what it currently knows. Learning history leaves a persistent, partially localizable trace in transient learner state, with w_{fast} providing a causal component of history-dependent future adaptation. This result gives a concrete mechanism for “development” in neural networks and clarifies a frequent but unsupported claim: developmental change does not equal universally faster learning.

Reproducibility

Code: <https://github.com/JayCRL/DLA> Experiment details: docs/experiments.md, paper/overnight_report.md, paper/validation_audit.md Scripts: stage4_formal.py, stage55e_wfast_p0.py, validation_p0_controls.py, validation_baselines.py, validation_second_setting.py, stage6_longitudinal.py, stage7_cross_domain.py, stage8_physics.py.

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